AIMS RIC — Doctoral Training School 2026

ML-ESS: A Multi-Agent AI Research Assistant

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Abstract

ML-ESS is a full-stack AI system that automates end-to-end research: given a natural language question, it orchestrates four specialised AI agents to search the web, synthesise evidence, draft a structured report, and evaluate the report's quality. The system exposes a REST API (FastAPI) and a modern web interface (Next.js), supporting real-time progress tracking, PDF export, and multi-provider LLM fallback. This report describes the system architecture, agent pipeline, data model, REST API, frontend application, deployment strategy, and design trade-offs.

Keywords: multi-agent systems, large language models, automated research, FastAPI, Next.js, retrieval-augmented generation.

1 Introduction & Motivation

Conducting thorough research manually is time-consuming: it involves querying multiple sources, extracting relevant claims, resolving contradictions, and composing a coherent, well-cited document. ML-ESS automates this entire workflow through a multi-agent pipeline, replacing hours of manual work with a structured, reproducible process.

Core objectives:

- Accept any research question in natural language.
- Autonomously gather evidence from the open web.
- Synthesise findings into themes, resolving contradictions.
- Generate a structured Markdown report with citations.
- Score the report quality along four dimensions.
- Deliver everything through a clean web UI with real-time feedback.

2 System Architecture

2.1 High-Level Overview

The system is divided into two main services: a Python backend responsible for all AI reasoning and data persistence, and a Next.js frontend that provides the user interface. Figure 1 illustrates the high-level architecture.

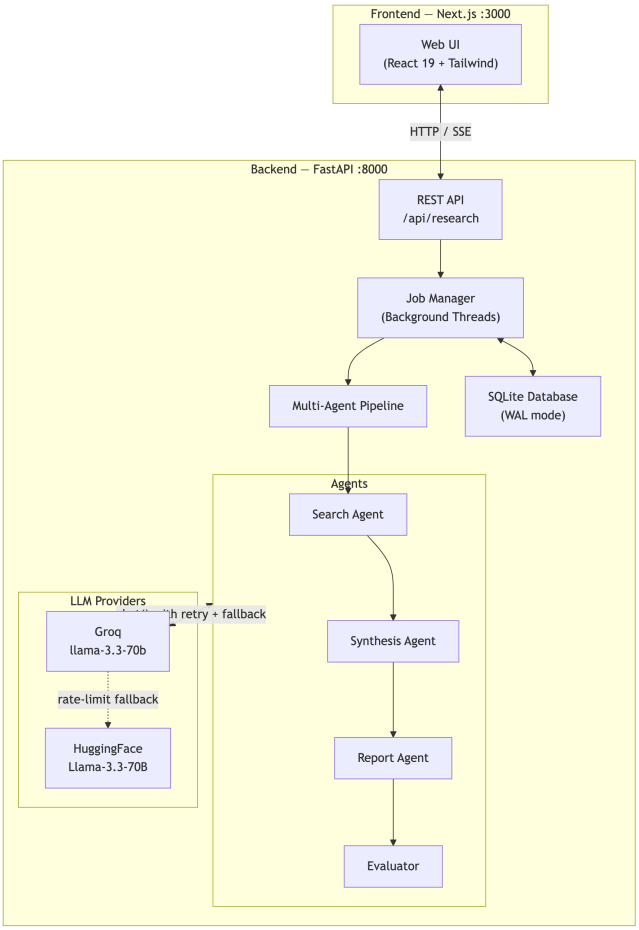


Figure 1: High-level system architecture.

2.2 Backend Directory Structure

```
api/
|-- main.py                 # CLI & server
    entry point
|-- app/
    |-- agents/
    |   |-- search.py       # Web search &
        evidence extraction
    |   |-- synthesis.py    # Theme
        identification
    |   |-- report.py       # Markdown
        generation
    |   |-- evaluator.py    # Quality scoring
    |-- api/
    |   |-- app.py          # FastAPI
        factory, CORS
```

```
|   |-- routes.py        # REST endpoints
    & SSE
|   |-- auth.py          # API key
    authentication
|-- core/
|   |-- pipeline.py      # Sequential
    orchestration
|   |-- jobs.py          # Background
    thread management
|   |-- store.py         # SQLite (WAL
    mode)
|   |-- llm.py           # LLM client w/
    retries
|-- models/
    |-- api.py           # Pydantic schemas
    |-- state.py         # Pipeline shared
        state
```

2.3 Request Lifecycle & Data Flow

Figure 2 shows the full request lifecycle from the moment a user submits a question to the delivery of the final report.

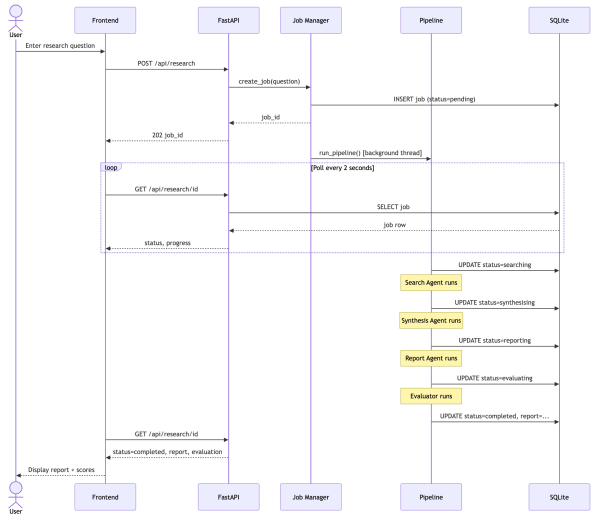


Figure 2: Request lifecycle and data flow (sequence diagram).

3 The Multi-Agent Pipeline

Each agent in the pipeline is a stateless function that reads from and writes to a shared `SharedState` object. Agents are called sequentially — each one enriches the state for the next.

3.1 SharedState Schema

Figure 3 shows the class diagram of the shared pipeline state, including the `Source`, `Evidence`, `Theme`, `Contradiction`, and `EvaluationScores` entities.

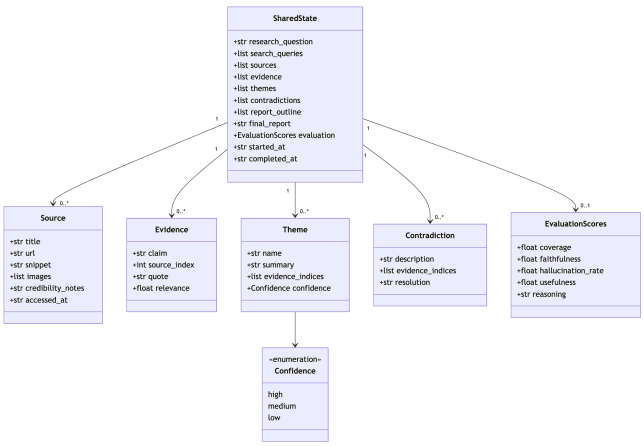
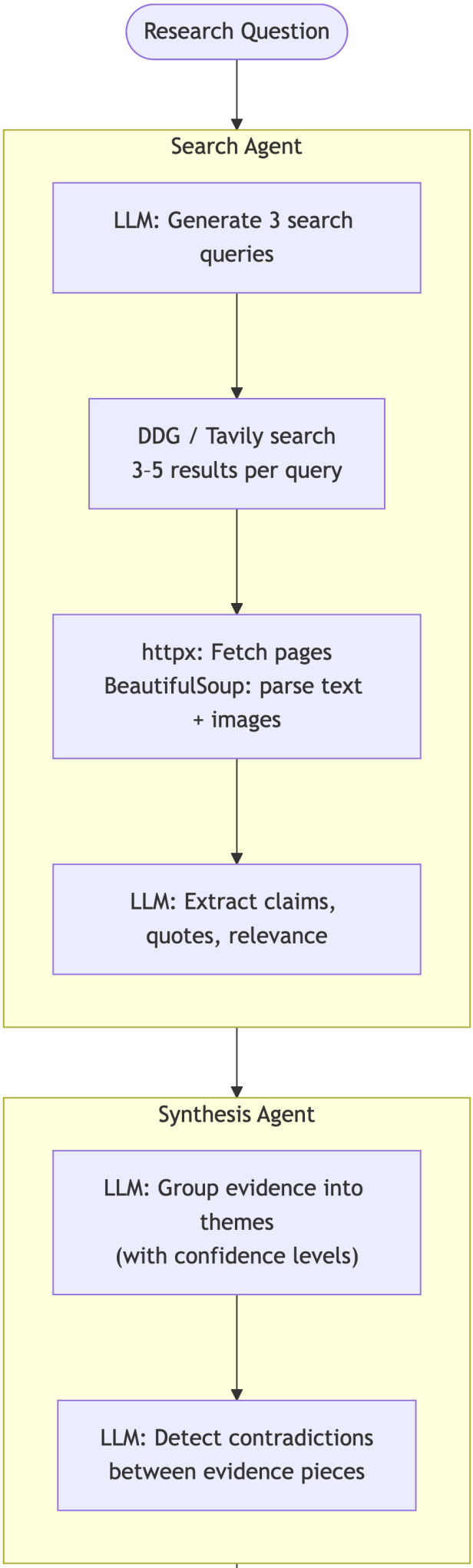


Figure 3: SharedState class diagram.

3.2 Agent Pipeline Flow

The pipeline executes four agents sequentially (Figure 4):

1. **Search Agent** — generates queries, fetches pages, and extracts structured evidence.
2. **Synthesis Agent** — groups evidence into themes and detects contradictions.
3. **Report Agent** — produces a structured Markdown report (~2 000–2 500 words) with citations.
4. **Evaluator** — scores the report on four quality dimensions.



3.3 Search Agent Detail

The Search Agent (Figure 5) generates up to three search queries via an LLM call, fans out to DuckDuckGo (or Tavily when an API key is provided), de-duplicates the result URLs, fetches page content with `httpx`, parses it with BeautifulSoup4, and finally calls the LLM again to extract structured evidence records.

Figure 5: Search Agent internal data flow.

3.4 LLM Provider Strategy

Figure 6 shows the retry and fallback strategy implemented in the LLM client. The primary provider is Groq (`llama-3.3-70b-versatile`); HuggingFace Inference (`Llama-3.3-70B-Instruct`) serves as a fallback when Groq rate-limits are exhausted.

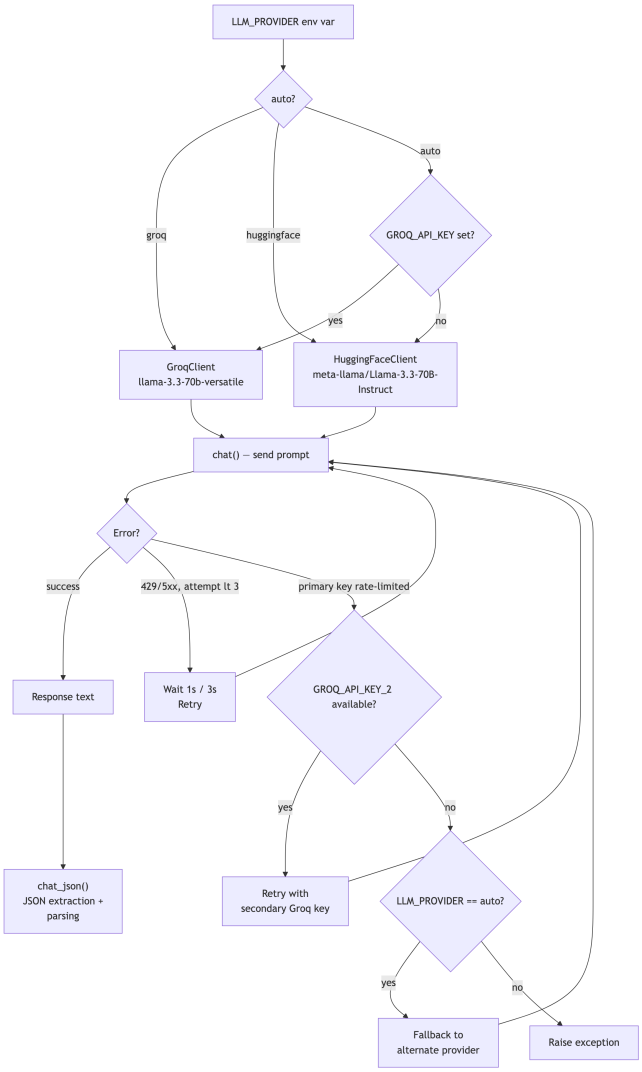


Figure 6: LLM provider retry and fallback strategy.

3.5 Evaluation Scores

The Evaluator Agent scores the generated report along four dimensions (Table 1). Scores for coverage, faithfulness, and usefulness are in $[0, 1]$; the hallucination rate is also in $[0, 1]$ but lower is better.

Table 1: Example evaluation scores.

Dimension	Score
Coverage	0.87
Faithfulness	0.92
Usefulness	0.85
Hallucination rate	0.08

4 Data Persistence

4.1 SQLite Schema

The backend persists all job data in a single SQLite database opened in WAL mode (Figure 7). Three tables are used: `JOBS` (primary record), `STATES` (full serialised pipeline state), and `EVENTS` (append-only SSE event log).

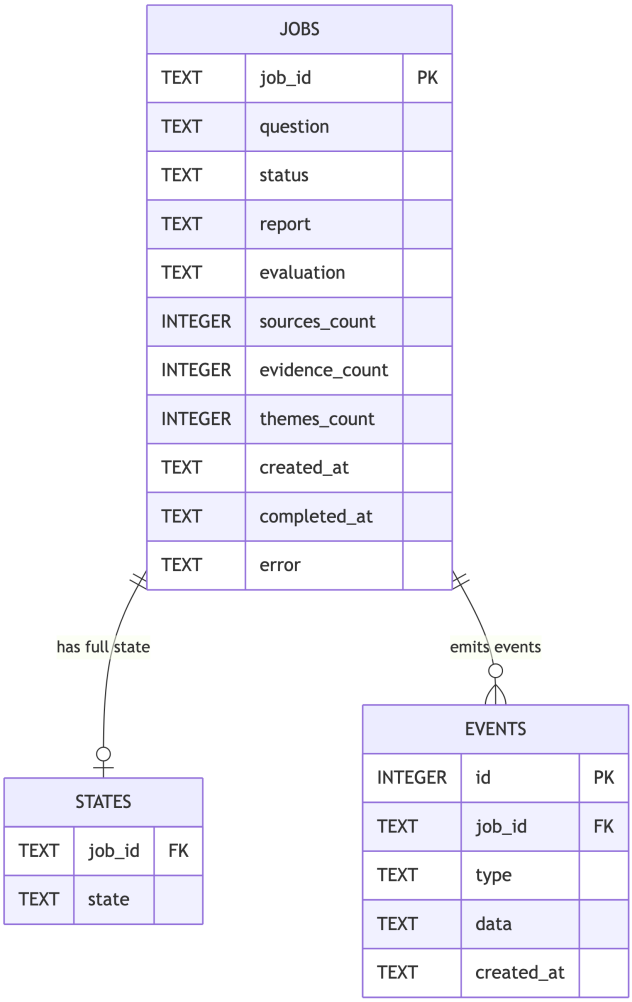


Figure 7: Entity-relationship diagram.

4.2 Job Status Lifecycle

Figure 8 shows the finite-state machine for a research job. A job progresses through *pending* → *searching* → *synthesising* → *reporting* → *evaluating* → *completed*, or transitions to *failed* on any unhandled exception.

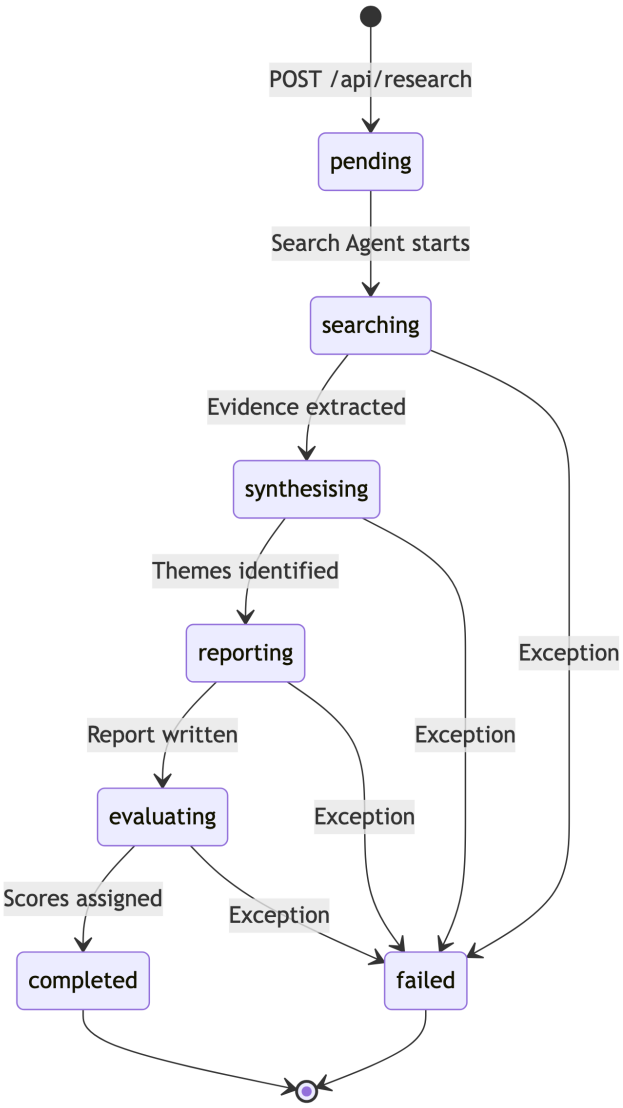


Figure 8: Job status state machine.

4.3 SSE Event Stream

Figure 9 shows the Server-Sent Events (SSE) protocol between the frontend and the `/api/research/{id}/events` endpoint.

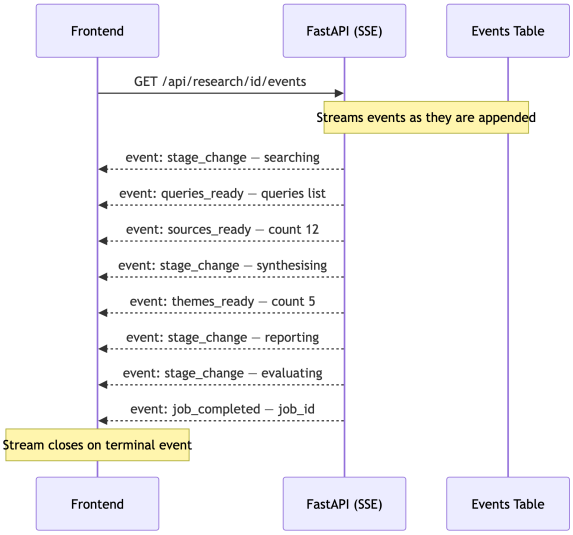


Figure 9: SSE event stream sequence.

5 REST API Reference

All endpoints except `/api/health` require the header `X-API-Key: <key>` when `API_KEY` is configured.

Table 2: REST API endpoints.

Method	Endpoint	Status
POST	/api/research	202
GET	/api/research	200
GET	/api/research/{id}	200
GET	.../{id}/reasoning	200
GET	.../{id}/events	200
GET	.../{id}/pdf	200
DELETE	/api/research/{id}	200
DELETE	/api/research	200
GET	/api/health	200

6 Frontend Application

The Next.js 16 frontend (React 19, Tailwind CSS v4, TypeScript) provides three pages: Home (/), Job History (`/history`), and Job Detail (`/job/[id]`). Figure 10 shows the page and component tree; Figure 11 shows the frontend data flow.

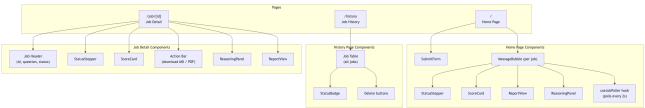


Figure 10: Frontend page and component tree.

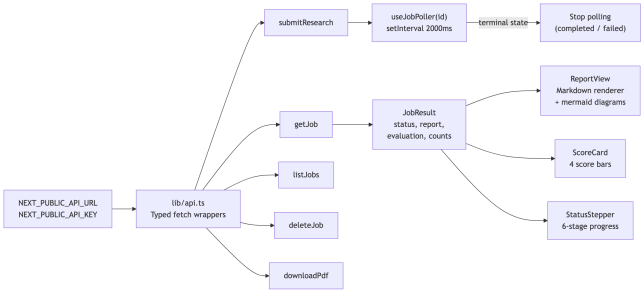


Figure 11: Frontend data flow.

7 Technology Stack

Table 3: Technology stack summary.

Layer	Technology	Role
Backend lang.	Python 3.11+	Agent & API logic
API framework	FastAPI ≥0.115	HTTP, routing
ASGI server	Uvicorn ≥0.30	Production server
LLM (primary)	Groq SDK	llama-3.3-70b
LLM (fallback)	HuggingFace Hub	Llama-3.3-70B
Web search	ddgs ≥9.0	DuckDuckGo
Web search (opt)	tavily-python	Premium search
Scraping	httpx + bs4	Fetch & parse
Validation	Pydantic ≥2	Schemas
Database	SQLite (WAL)	Persistence
PDF export	WeasyPrint ≥62	MD → PDF
Frontend	Next.js 16.2.1	React 19 App Router
Styling	Tailwind CSS v4	Utility-first CSS
Language (FE)	TypeScript	Type safety
Deployment (FE)	Vercel	Hosted frontend

8 Configuration

Key environment variables (set in `api/.env`) are listed in Table 4.

Table 4: Key environment variables.

Variable	Default	Purpose
`LLM_PROVIDER`	auto	groq / hf / auto
`GROQ_API_KEY`	—	Primary Groq key
`GROQ_API_KEY_2`	—	Fallback Groq key
`GROQ_MODEL`	llama-3.3-70b	Groq model
`HF_TOKEN`	—	HuggingFace token
`TEMPERATURE`	0.3	LLM temperature
`TAVILY_API_KEY`	—	Premium search
`MAX_SEARCH_QUERIES`	3	Queries / question
`MAX_RESULTS_PER_QUERY`	5	Results / query
`MAX_CONTENT_LENGTH`	3000	Chars / page
`API_KEY`	—	Auth key
`DB_PATH`	data/jobs.db	SQLite path
`CORS_ORIGINS`	Vercel URL	Allowed origins

9 Deployment Architecture

9.1 Current vs. Scalable Architecture

Figure 12 contrasts the current single-instance deployment (FastAPI + SQLite + background threads on a single process) with a production-scale horizontal architecture (load balancer, multiple FastAPI workers, Redis job queue, Celery workers, and PostgreSQL).

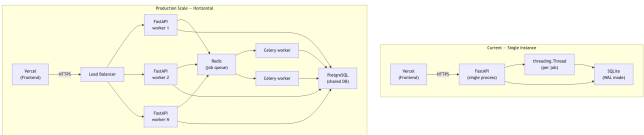
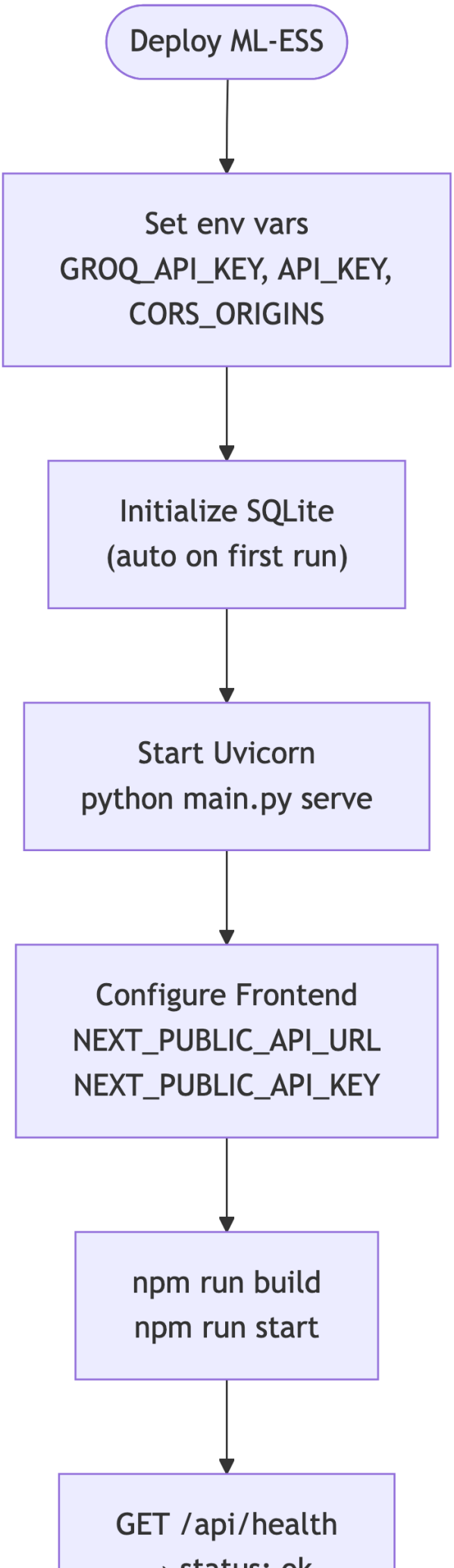


Figure 12: Current vs. scalable deployment architecture.

9.2 Deployment Checklist

Figure 13 summarises the five-step deployment procedure: set environment variables, initialise SQLite, start Uvicorn, configure the frontend, and verify the health endpoint.



10 Related Work

Autonomous agent frameworks. AutoGPT [7] and BabyAGI [8] pioneered open-ended LLM agents capable of self-directed task decomposition, but they lack a structured evidence-gathering and quality-scoring loop. CrewAI [9] introduces role-based multi-agent crews; ML-ESS differs by using a strict linear pipeline where each stage produces typed artefacts consumed by the next, making the data flow auditable and reproducible.

Retrieval-augmented generation. Lewis *et al.* [5] established that grounding LLM generation in retrieved passages substantially reduces hallucination. ML-ESS extends this pattern: the Search Agent plays the role of the retriever, and the Report Agent is conditioned on structured `Evidence` records rather than raw retrieved text, enabling the Evaluator to score faithfulness precisely.

LLM orchestration libraries. LangChain [10] and LlamaIndex provide agent primitives and retrieval pipelines. ML-ESS intentionally avoids these dependencies: the entire pipeline is implemented in ~600 lines of plain Python, keeping the system transparent, easy to audit, and free of framework-specific abstractions.

Commercial research assistants. Perplexity AI and Elicit offer similar research-assistant experiences but are closed-source and provide no quality scores or structured intermediate state. ML-ESS is fully open, self-hostable, and returns the complete `SharedState` for downstream inspection.

11 Design Decisions & Trade-offs

11.1 Sequential vs. Parallel Agents

Agents run sequentially because each stage produces the typed artefacts that the next stage requires as input: the Synthesis Agent cannot run before `sources` and `evidence` are populated; the Report Agent cannot run before `themes` exist. Parallelism would require speculative execution and complex dependency tracking for negligible gain given that the bottleneck is LLM latency, not CPU.

11.2 SQLite over PostgreSQL

For a single-process deployment serving one research job at a time, SQLite in WAL mode provides adequate write throughput with zero operational overhead (no separate database process, no connection pooling). The database schema is designed so that all writes during a pipeline run are sequential updates to a single row — a pattern SQLite handles efficiently. The scalable architecture diagram (Figure 12) shows the migration path to PostgreSQL when horizontal scaling is needed.

11.3 Threading over Celery

Background jobs use `threading.Thread` rather than a Celery/Redis worker queue. This avoids adding two additional services (a message broker and worker processes) to the deployment footprint, which is appropriate for a research prototype. The trade-off is that jobs cannot survive a process restart and cannot scale across multiple machines without the architectural changes shown in Section 9.

11.4 LLM Temperature at 0.3

A temperature of 0.3 is deliberately low. Research synthesis requires factual, reproducible outputs: query generation should produce consistent search terms for a given question, and evidence extraction should classify claims deterministically. A higher temperature introduces variance that makes the pipeline harder to debug and evaluate.

11.5 Dual LLM Provider

Groq's inference API offers sub-second token generation via custom LPU hardware, making it the primary provider. Its free tier has strict per-minute rate limits. The HuggingFace Inference API serves as a fallback: slower, but with a separate rate-limit bucket. The `LLM_PROVIDER=auto` strategy automatically selects and fails over between them, insulating the pipeline from either provider's availability.

12 Security Considerations

Authentication. All API endpoints except `/api/health` are protected by a static API key passed in the `X-API-Key` header. When `API_KEY` is empty (the default in development), authentication is disabled. Production deployments must set a strong random key.

CORS. The FastAPI application enforces a strict `CORS_ORIGINS` allowlist. Only the configured frontend origin (typically the Vercel deployment URL) may make cross-origin requests, preventing credential leakage to third-party pages.

Input handling. Research questions are passed verbatim to LLM prompts. There is currently no sanitisation layer against prompt-injection attacks — a hostile question could attempt to override system instructions. A hardened deployment should prepend an untrusted-input envelope and validate that the LLM's JSON responses conform to the Pydantic schema (which the pipeline already does for structured responses).

Secrets management. API keys are loaded from a `.env` file via `python-dotenv`. That file must not be committed to version control; a `.gitignore` entry is present in the repository.

13 Limitations & Future Work

13.1 Current Limitations

- **No cross-job memory.** Each research job is independent; the system cannot reuse evidence gathered in a previous run on a related question.
- **LLM-graded evaluation.** The Evaluator scores the report using the same LLM that produced it, introducing a potential circular bias. Human evaluation or a separate judge model would be more reliable.
- **Synchronous web scraping.** Pages are fetched sequentially with `httpx`; slow servers can add tens of seconds to the Search Agent phase.
- **No source credibility verification.** The credibility notes attached to each `Source` are LLM-generated heuristics, not verified signals (domain authority, publication date, peer review status).
- **SQLite concurrency ceiling.** WAL mode supports multiple concurrent readers but only one writer. Submitting many jobs simultaneously will queue write operations; under sustained load this becomes a bottleneck.

13.2 Future Directions

- **Streaming LLM output.** Exposing token-by-token SSE events from the Report Agent would allow the frontend to display the report as it is being written, dramatically improving perceived latency.
- **Parallel scraping.** Replacing sequential `httpx` calls with `asyncio.gather` could cut Search Agent latency by 3×.
- **Cross-job knowledge base.** A persistent vector store (e.g. ChromaDB, pgvector) would allow the Synthesis Agent to retrieve evidence from previous runs, enabling incremental research.
- **Human-in-the-loop feedback.** A researcher could annotate or reject individual `Evidence` records before the Report Agent runs, improving output quality for high-stakes use cases.
- **Independent evaluator model.** Using a second, independent LLM (or a specialised scoring model) as the Evaluator would eliminate the self-grading bias noted above.
- **Multi-language support.** Adding a translation agent at the output stage would make the system accessible to non-English researchers.

14 Conclusion

ML-ESS demonstrates how a small, well-structured multi-agent system can automate a complex, knowledge-intensive task end-to-end. By composing four focused agents — each with a clear role — the pipeline transforms a single natural language question into a cited, evaluated research report in minutes.

The system is designed for clarity and correctness: low LLM temperature for deterministic outputs, retry/fallback logic for resilience, persistent job state for observability, and a clean typed API for frontend integration. The modular agent design makes the system straightforward to extend: adding a new research phase (e.g., a fact-checking agent or a translation layer) requires only inserting a new agent function into the pipeline and extending `SharedState` with the new fields.

Acknowledgements

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